

**Question:**

Carla is downloading a 200 GB file. Normally she can download 2 GB/minute, but 40% of the way through the download, Windows forces a restart to install updates, which takes 20 minutes. Then Carla has to restart the download from the beginning. How long does it take to download the file?

**Reasoning + Answer:**

Let's solve this step by step :

- 1. **Total file size**: 200 GB
- 2. **Normal download speed**: 2 GB / minute
- 3. **40 % of the way through**:
  - 40 % of 200 GB =  $0.40 \times 200 = 80$  GB
  - So , Carla downloads 80 GB before the restart .

- 4. **Time to download 80 GB at 2 GB / minute**:
  - Time = Size / Speed =  $80 / 2 = 40$  minutes

- 5. **Restart due to update**:
  - Takes 20 minutes

- 6. **After restart , she has to start over**:
  - She needs to download the entire file again from the beginning , so she downloads 200 GB again at 2 GB / minute
  - Time =  $200 / 2 = 100$  minutes

- 7. **Total time** = time before restart + restart time + time after restart  
=  $40 + 20 + 100 = 160$  minutes

So , the total time it takes to download the file is 160 minutes .

< answer > 160 </ answer >

